

where α_t is the trading rate that will be used from the current time through the completion of the order.

Target Cost Tactic

The targeted cost adaptation tactic will minimize the squared difference between the projected cost and original cost from the best execution strategy. This tactic will always revise the strategy to put us back on track to get as close as we can to the original expected trading cost. Here, the strategy becomes more aggressive in times of favorable price movement and more passive in times of adverse price movement.

Mathematically, the target cost optimization is found by minimizing the following:

$$\begin{aligned} \text{Min } L &= (E_0(\text{Cost}) - E_t(\text{Cost}))^2 \\ \text{s.t. } LB^* &\leq \alpha_t \leq UB^* \end{aligned}$$

The general optimization needs to include expectations for volume and volatility over the remainder of the trading period. Volatility can be adjusted by the HMA-VIX adjustment described in Chapter 6, Price Volatility. The upper and lower bounds are included to ensure that the optimal strategy will be within levels specified by the investor. Some investors request to trade no slower than some level (e.g., $\geq 5\%$) and no faster than another level (e.g., $\leq 40\%$). Furthermore, most investors do require completion of the order by some end time. Thus the optimal strategy needs to be at least fast enough to ensure that trading will be completed by the specified end time or market close.

For the trading rate strategy the targeted cost adaptation tactic objective function is:

$$\text{Min } L = (C^* - (K_t + (1 - \theta)(b_1 \cdot I' \cdot \alpha_t)))^2 \quad (8.35)$$

where C^* is the original expected cost from the BES, and K_t is the unavoidable cost at time t based on realized cost and market movement. Solving for α_t we get,

$$\alpha_t = \frac{C^* - K_t}{(1 - \theta)(b_1 \cdot I')} \quad (8.36)$$

This rate is then adjusted to ensure it satisfies our boundary conditions. That is written mathematically as:

$$\alpha_t^* = \min(\max(LB, \alpha_t), UB)$$

Figure 8.3a compares the cost distribution from a targeted cost adaptation tactic to the cost distribution of a constant trading rate. The expected cost of the targeted rate C_2 will be lower than the constant rate C_1 since this tactic